Zijie Zhou

CONTACT INFORMATION

College of Artificial Intelligence
China University of Petroleum (Beijing)

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EDUCATION

China University of Petroleum (Beijing) Beijing, China

Master of Engineering in Artificial Intelligence *Aug. 2023 - Jun. 2026*

- Average Grade: 90.20/100 | Rank: 3/31
- Relevant Coursework: Methods of Optimization, Introduction to Artificial Intelligence, Introductory Combinatorics
- Dissertation Topic: "Research on Graph-Based Collaboration Mechanisms for Multimodal Multi-Agent Systems"

China University of Petroleum (Beijing) Beijing, China

Bachelor of Engineering in Artificial Intelligence *Aug. 2019 - Jun. 2023*

- Average Grade: 86.12/100 | Rank: 5/33
- Relevant Coursework: Machine Learning, Reinforcement Learning, Fundamentals of Deep Learning, Python

PUBLICATIONS

Zijie Zhou, Dandan Zhu, Hangxiangpan Wang, Heng Zhang, Huishen Jiao, Yi Zhao. *Hyperbolic and Evidence-Prioritized Experts for Large Vision-Language Models.* ICMR 2026 (Best Paper Candidate). [Paper]

Zijie Zhou, Dandan Zhu, Heng Zhang, Yanming Yang, Xu Liu, Huishen Jiao. *Formalizing Heterogeneous Interaction Modeling for Multi-Agent Debate in Visual Geo-Localization.* ICME 2026 (accepted; proceedings forthcoming). [Paper]

Jianyu Wang*, **Zijie Zhou***, Wenjie Shao, Xi Xiao, Duoxun Tang, Qing Li. *From Capability to Efficiency: Learning to Orchestrate LLM Routing via Progressive Reinforcement Learning.* Under Review, AAAI 2027.

Xiangxiang Meng, **Zijie Zhou**. *LogicVLA: Neural Logic Critics for Efficient Vision-Language-Action Fine-Tuning.* Under Review, ICLR 2027.

Taolin Zhang*, Weizi Shao*, **Zijie Zhou**, Chen Chen, Daiyang Yu, Tingyuan Hu, Chengyu Wang, Xiaofeng He. *M^3Prune: Hierarchical Collaborative Pruning for Efficient Multi-Modal Multi-Agent Retrieval-Augmented Generation.* ACM MM 2026. [Paper].

Yuqi Li, **Zijie Zhou**, Zhiyuan Peng, Junhao Dong, Haochen You, Renye Yan, Shiping Wen, Yingli Tian, Tingwen Huang. *A Preference-Driven Methodology for Efficient Code Generation.* IEEE Transactions on Artificial Intelligence, 2025. [Paper]

ACADEMIC EXPERIENCE

Tsinghua Shenzhen International Graduate School Shenzhen, China

Research Collaborator

EvoRouter: Hierarchical LLM Routing via Reinforcement Learning *Apr. 2025 - Sep. 2025*

- Contributed the core research idea for EvoRouter, proposing reinforcement learning for dynamic multi-turn LLM routing to balance task performance, latency, and token cost.
- Designed the experimental evaluation plan, including benchmark selection, baseline comparisons, and ablation studies.

China University of Petroleum (Beijing) Beijing, China

Master's Thesis — Research on Graph-Based Collaboration Mechanisms for Multimodal Multi-Agent Systems *Dec. 2024 - Apr. 2026*

- Developed a meta-learning framework integrating conditional variational graph generation and domain-adversarial learning to construct adaptive collaboration topologies, improving cross-domain adaptation by 2.87% while reducing token consumption by 40.22%.
- Proposed a heterogeneous graph-based debate framework with dual-level reasoning and adaptive topology refinement, improving visual geo-localization accuracy by up to 2.47% over the strongest baseline.
- Designed a hierarchical graph pruning framework integrating intra-modal and inter-modal sparsification with progressive edge pruning, improving accuracy by 9.4% while reducing token consumption by 15.7%.

Peking University Beijing, China

Research Collaborator

Efficient Code Generation via Reinforcement Learning *Dec. 2024 - Mar. 2025*

- Designed a composite reward function incorporating compilation success, functional correctness, and execution efficiency for reinforcement learning-based code generation.
- Designed comparative experiments to evaluate the proposed method against baseline approaches in terms of code correctness and execution efficiency.

Working Experience

Innovation Business Center of Yidu Cloud Beijing, China

Algorithm Intern *Feb. 2023 - Aug. 2023*

Developed GRU-based multi-label models for disease prediction with competitive Top-10 performance; cleaned and linked internal medical records via SQL and Pandas to enable effective downstream business analysis.

PERCENT Technology, International Business Department Beijing, China

Machine Learning Research Assistant *Aug. 2022 - Feb. 2023*

Improved Chinese-to-English machine translation by 2%, delivering the deployed model; additionally led multilingual data standardization and contributed to sentiment classification system design.

Teaching Experience

Principle of Artificial Intelligence *Sep. 2024 - Nov. 2024*

Teaching Assistant, B.Sc AI course in CUPB Beijing, China

Python for Data Analysis *Apr. 2024 - Jun. 2024*

Teaching Assistant, B.Sc course in CUPB Beijing, China

Honors and Awards

Graduate Study Scholarship, 2023

Undergraduate Study Scholarship, 2022

Technology Innovation Advanced Individual Award, 2022 and 2023

Honorable Mention, Interdisciplinary Contest In Modeling, 2021

Outstanding Undergraduate Student Cadre, 2020

Technical Skills

- Programming: Python, PyTorch
- Research: Reinforcement Learning, Multi-Agent Systems, LLM Fine-Tuning, RLHF
- Tools: Git, Docker, Linux, LaTeX, Hugging Face, Weights & Biases